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SECP3213 BUSINESS INTELLIGENCE

SECTION 02

Project Proposal

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1.0 Project Title

Comparative Sales Analytics and Weather-Driven Demand: Optimizing Restaurant Performance Using Business Intelligence

2.0 Problem Background

2.1 Business Context

The food and drink sector is a fiercely competitive one, and it's important that restaurants continuously assess their business and adjust their strategies to meet their customers' evolving habits. For management that has more than one branch (for instance, Restaurant 1 and Restaurant 2), it can be difficult to get a complete overview of their performance. In addition, other external environmental factors significantly affect customer purchasing behavior; weather (temperature, precipitation, and cloudiness) is a key factor determining the daily demand. Today, these restaurant decision-makers do not have a consistent interactive benchmarking system among restaurants to evaluate performance across branches and understand what influences restaurant sales.

2.2 Importance of the Problem

If there were no business intelligence strategy, it would be very hard to analyze raw transaction data from various branches and external weather logs. This poses various business challenges:

- No Performance Benchmarking: Without easy comparison of performance, it is hard for management to determine why one restaurant may be more successful than another in terms of revenue, average order value, and product popularity, thus making it difficult to standardize best practices.
- Weather has an impact: If you don't know how the weather affects sales, you risk having the wrong inventory and staff available. For example, when the weather is rainy, there may be more orders from customers, but if it's hot, there might be a change in demand for certain menu items – in either case, food will be wasted and revenue opportunities missed.
- Inefficient Pricing and Menu Strategy: Without knowing the correlation between the product price, order quantity and the revenue contribution (80/20 rule), the restaurants will not be able to optimize their Menus for maximum profit.

2.3 Expected Impact

This project will provide insights that can be acted upon by combining the transaction information from the two restaurants with existing London weather data and creating interactive dashboards using Power BI or Tableau after preprocessing the data in Alteryx Designer. The potential effects will be:

1. Data-Driven Branch Optimization: Using KPI dashboards, management will be able to directly compare Restaurant 1 and 2, and find out overall levels of efficiency and apply the good ideas of the winning product strategy to other restaurants.
2. Weather-Responsive Operations: The temperature and precipitation effects are displayed as heat maps and scatter plots, enabling the restaurants to make suitable changes in inventory and employee schedules in advance according to the weather forecast.
3. By leveraging seasonal trends and peak ordering periods, Strategic Menu Engineering: Insights will enable the business to implement targeted marketing campaigns and adjust pricing strategies to better align with customer demand.

3.0 Objective

The objectives of this project are as follows:

1. To develop a business intelligence solution by integrating food order details, product pricing and weather datasets using Alteryx Designer.
2. To compare and analyse the sales performance, customer ordering behaviour and product popularity between Restaurant 1 and Restaurant 2.
3. To examine the impact of weather conditions on restaurant sales and customer demand.
4. To create interactive dashboards and visualizations using Power BI to support data-driven decision making and performance evaluation.

4.0 Datasets Description

1. **Dataset 1: Restaurant 1 Orders**
   Source: Kaggle-Indian Takeaway Restaurant 1 Orders in London Dataset
   Type: Structured data (CSV format)
   Description:
   This dataset contains customer takeaway order records from an Indian restaurant, namely Restaurant 1 in London, United Kingdom. Each row represents a product ordered by a customer, including order details, product name and prices, quantity purchased and total product amount in the same order number.

   Key Variables

   - Order Number
   - Order Date
   - Item Name
   - Quantity
   - Product Price
   - Total Product

2. **Dataset 2: Restaurant 1 Product Prices**

   Source: [Kaggle-Indian Takeaway Restaurant Prices Dataset]()
   Type: Structured data (CSV format)
   Description:
   This dataset contains all menu items and pricing information for Restaurant 1.

   Key Variables

   - Item Name
   - Product Price

3. **Dataset 3: Restaurant 2 Orders**

   Source: [Kaggle-Indian Takeaway Restaurant 2 Orders Dataset]()
   Type: Structured data (CSV format)
   Description:
   This dataset contains customer takeaway order records from an Indian restaurant, namely Restaurant 2 in London, United Kingdom. Each row represents a product ordered by a customer, including order details, product name and prices, quantity purchased and total product amount in the same order number.

   Key Variables

   - Order ID
   - Order Date
   - Item Name
   - Quantity
   - Product Price
   - Total Product

4. **Dataset 4: Restaurant 2 Product Prices**

   Source: [Kaggle-Indian Takeaway Restaurant 2 Prices Dataset]()
   Type: Structured data (CSV format)
   Description:
   This dataset contains all menu items and pricing information for Restaurant 2.

   Key Variables

- Item Name
- Product Price

5. **Dataset 5: London Weather Dataset**

   Source: [Kaggle-London Weather Data]

   Type: Structured data (CSV format)
   Description:
   This dataset contains historical weather information in London, including temperature, precipitation, sunshine and cloud cover measurements from 1979 to 2020, which can be used to analyse the impact of weather conditions on customer food ordering behaviours.

   Key Variables

   - date
   - cloud cover
   - sunshine
   - global_radiation
   - max_temp
   - mean_temp
   - precipitation
   - pressure
   - snow_depth

5.0 Proposed BI Approach

5.1 Alteryx and visualization tools usage

In this project, Alteryx Designer will be used to prepare and process the datasets before analysis. Alteryx provides a drag-and-drop ETL workflow environment that allows users to automate data preparation processes efficiently. The tool will be used to blend multiple datasets, perform automated preprocessing tasks such as filtering, joining, cleansing, and transforming data, and create repeatable workflows for consistent analysis.This approach improves data quality and reduces manual processing time. The datasets from Restaurant 1, Restaurant 2, and the London weather dataset will be imported into Alteryx for preprocessing and workflow management. The tool will also help organize the data into a structured format suitable for Business Intelligence analysis.

After the datasets are prepared, Power BI or Tableau will be used to create interactive dashboards and visual reports. These visualization tools enable users to present business information in a clearer and more understandable format through charts, graphs, and dashboard interfaces. The dashboards will help users monitor restaurant performance, observe sales trends, and support business decision-making through visual analytics.

Overall, the integration of Alteryx and visualization tools provides a complete Business Intelligence solution that transforms raw data into meaningful and interactive business insights.

5.2 Data Integration, Data Cleaning and Data Transformation Methods

5.2.1 Data Integration

The project integrates five datasets consisting of restaurant transaction records, product pricing data, and London weather information. The Restaurant 1 Orders dataset will be integrated with the Restaurant 1 Product Prices dataset using the item name attribute, while Restaurant 2 Orders will be combined with Restaurant 2 Product Prices in the same manner. After that, both restaurant datasets will be merged to enable cross-branch performance comparison. The London weather dataset will then be integrated with the restaurant transaction datasets based on the order date to analyze the influence of weather conditions on restaurant sales and customer purchasing behaviour.

5.2.2 Data Cleaning

Several data cleaning methods will be performed to ensure data quality and consistency before analysis. Missing values and duplicate records will be identified and removed where necessary. Inconsistent naming formats for menu items and date fields will also be standardized to ensure compatibility during dataset integration. Invalid or incomplete transaction records will be filtered out to improve analysis accuracy.

5.2.3 Data Transformation

Data transformation techniques will be applied to prepare the datasets for Business Intelligence analysis. New calculated fields such as total revenue and average order value will be created to support KPI analysis. Date fields will be transformed into monthly, seasonal, and weekday categories to enable trend analysis. Aggregation methods will also be used to summarize sales performance, product demand, and weather-related ordering patterns.

5.3 Planned Analysis and Proposed Visualizations

Several analyses will be conducted to provide meaningful business insights for restaurant management. Firstly, restaurant performance comparison analysis will be conducted to compare total sales, average order value, and order volume between Restaurant 1 and Restaurant 2. Secondly, weather impact analysis will be performed to identify how weather conditions such as precipitation and temperature influence customer purchasing behaviour and restaurant sales.

In addition, seasonal sales trend analysis will be conducted to identify peak ordering periods and seasonal demand patterns. Product popularity and revenue contribution analysis will also be performed to determine the best-selling and highest revenue-generating menu items. Furthermore, price versus demand analysis will be conducted to examine the relationship between product pricing and customer demand.

To support these analyses, several appropriate visualization forms will be used:

- **Line charts** to display sales trends over time
- **Bar charts** to compare restaurant performance and top-selling products
- **Scatter plots** to analyze the correlation between weather conditions and sales
- **Heatmaps** to visualize weather impact on customer demand
- **KPI dashboards** to summarize key business indicators such as total revenue, average order value, and total orders

Overall, the proposed BI solution will support data-driven decision making in areas such as branch benchmarking, inventory planning, staffing management, menu optimization, and targeted marketing strategies.